

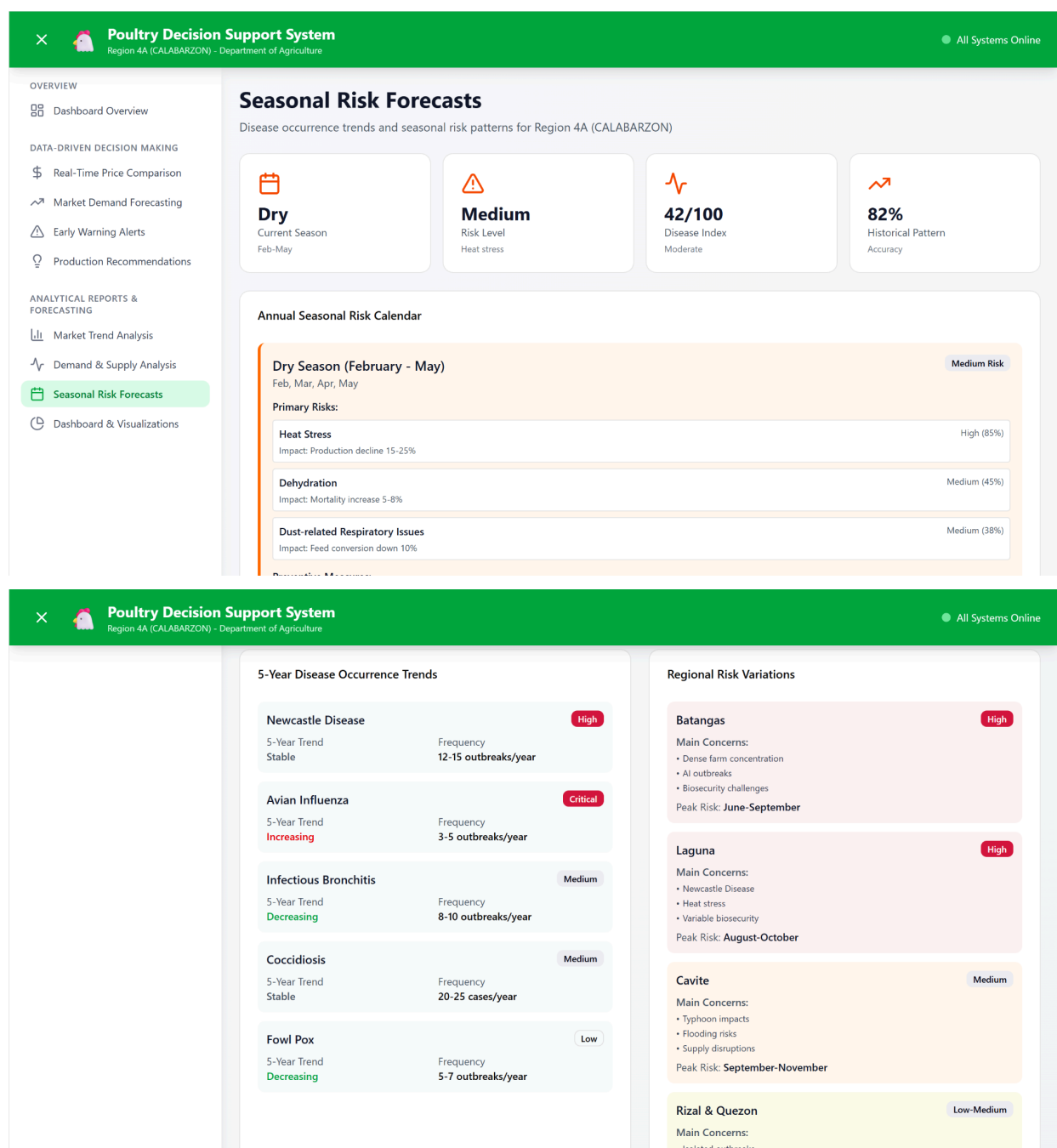
Subsystem Prototype



Dashboard & Visualizations

Projected Growth

Production Recommendations Based on Forecast



The three subsystem prototypes embody the fundamental functionalities of the envisioned Poultry Decision Support System, each tailored to assist both poultry farmers and the Department of Agriculture in Region IV-A. These subsystems are integrated to furnish prompt market intelligence, forecast potential risks, and formulate well-informed recommendations, thereby facilitating improved decision-making processes.

The first subsystem, Supply Chain and Price Monitoring, is dedicated to monitoring current market dynamics and conducting price comparisons across various geographical areas.

This module allows stakeholders to view current price levels, market differences, price trends, and possible arbitrage opportunities. By consolidating supply chain and pricing data in a single interface, the subsystem helps farmers identify potentially more profitable sales opportunities and allows the DA to monitor market changes more effectively.

The second subsystem, Risk Forecasting and Early Warning, is designed to identify potential threats before they significantly harm poultry production. This includes monitoring disease outbreaks, seasonal risks, and other environmental factors that could disrupt farming operations. The module presents alert levels, regional risk status, and seasonal disease patterns, while also recommending preventive actions such as biosecurity measures and mitigation strategies. This feature provides stakeholders with advanced notice, potentially allowing them to mitigate losses and bolster their readiness.

The Decision Support Dashboard, the system's third subsystem, serves as its central analytical and advisory component. This element synthesizes data from market, supply chain, and risk forecasting modules to generate user-specific, actionable insights. Consequently, the dashboard offers production recommendations, trend analyses, and performance metrics, thereby assisting farmers in optimizing production levels, managing feed consumption, and improving operational planning. This subsystem effectively converts raw data into practical guidance, thereby facilitating both immediate and strategic decision-making processes.

Conceptual Framework

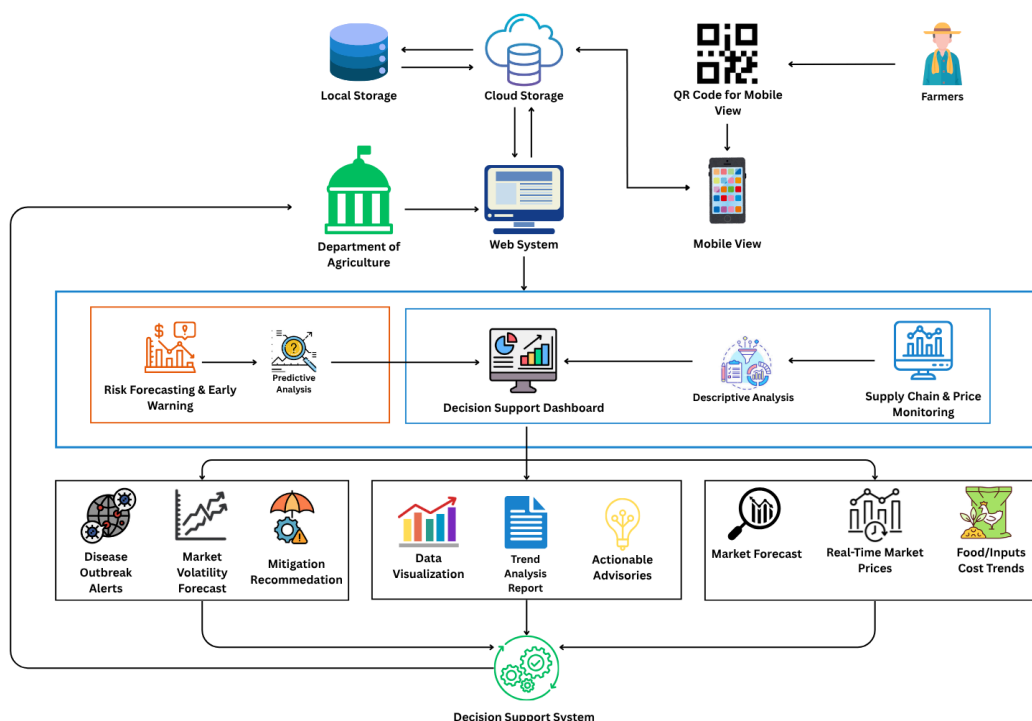


Figure 1.1 Conceptual Framework of Proposed Study

The conceptual framework illustrates the cyclical data flow and architecture of the proposed Decision Support System and Risk Forecasting tool designed for poultry and egg stakeholders and the Department of Agriculture (DA) in Region IV-A. The process begins when farmers access the system by scanning a QR code, allowing them to input real-time farm data such as production levels, feed consumption, environmental status, and mortality rates. This information is instantly transmitted through a secure cloud database—which is also synced to local storage for backup—directly to the DA's web system. Acting as the central administrator, the DA combines the farmers' information with macro-level inputs, including regional policies, market and supply chain statistics, reported disease outbreaks, and environmental changes. The DA then processes this integrated data using three core subsystems: Decision Support, Risk Forecasting & Early Warning, and Supply Chain Monitoring. To analyze the data, the system relies on descriptive analytics using a Simple Moving Average (SMA) approach to smooth out short-term fluctuations and identify clear historical trends in egg prices, feed costs, and production levels. Furthermore, the system employs Time-Series analysis for predictive analytics; by examining time-based data and seasonal variations, the Time-Series models can accurately forecast future market dynamics, impending disease outbreaks, and the potential impacts of environmental shifts. Once the information is processed, the system completes the communication loop by sending targeted, actionable insights back to the farmers' mobile view. Farmers receive real-time price updates, revenue profitability metrics, and a visual dashboard tracking feed efficiency and poultry loss, along with critical disease and climate alerts. Simultaneously, the DA receives aggregated regional market analytics and disease outbreak forecasts, empowering them to proactively generate and disseminate data-driven advisories to help farmers mitigate ongoing agricultural challenges.

Conclusion

The conceptual framework shows that farmers and the Department of Agriculture share a lot of information with each other over and over again. On their phones, farmers enter real-time information about their farms. The DA then puts this data together with data on local markets, reports of diseases, and information about the environment. The three main parts of the system then work together to handle the data: the Decision Support Dashboard, Risk Forecasting and Early Warning, and Supply Chain and Price Monitoring. This process leads to the ideas and suggestions that can be put into action.

It is clear from the framework that the proposed system is meant to do more than just record data. It is also to look at trends, predict risks, and help the poultry and egg business make decisions. The method can help people understand how the market changes, predict problems, and act more effectively by using descriptive analytics and Time-Series analysis.